The Math Behind CPA Uncertainty

A step-by-step study guide with fully worked examples

Written to accompany the CPA partial-derivative project

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How to use this guide

This guide teaches, from the ground up, every piece of math needed to understand how the CPA (Closest Point of Approach) uncertainties are computed and why the algebraic partial derivatives work. Nothing is skipped: every example shows the arithmetic. If a section is already familiar, skim its worked example and move on.

The big picture in one paragraph. Two objects move. We describe their *relative* motion with a position $\mathbf{p}$, velocity $\mathbf{v}$ and acceleration $\mathbf{a}$. The CPA time t^* is when they are closest. At that moment we read off the range and bearing. Our measurements of $\mathbf{p}, \mathbf{v}, \mathbf{a}$ are uncertain, so we ask: how uncertain are t^*, the range and the bearing? The answer needs (i) *partial derivatives*, which tell us how each output reacts to each input, and (ii) *error propagation*, which combines those sensitivities with the input uncertainties.

Part	Topic	Why it is needed
1	Vectors, dot product, bearing	Positions, velocities, "perpendicular"
2	Motion and relative motion	Where the formula $\mathbf{R}(t) = \mathbf{p} + \mathbf{v}t + \frac{1}{2}\mathbf{a}t^2$ comes from
3	Derivative rules	Differentiating $\mathbf{R}\cdot\mathbf{R}$ and $\mathbf{R}\cdot\mathbf{V}$
4	Partial derivatives and Jacobians	Sensitivities of outputs to inputs
5	CPA with constant velocity	The simple case, solved completely
6	CPA with acceleration	The cubic equation and Newton's method
7	Implicit differentiation	The key trick: derivative of a root
8	Range and bearing derivatives	Finishing the Jacobian
9	Error propagation	Turning a Jacobian into standard deviations
10	The full scenario, end to end	Every number in one place
11	Numerical differentiation	Why the step size matters
12	Practice problems	With complete solutions
13	Math-to-MATLAB dictionary	Connecting symbols to code

Notation. Bold letters are 2-component vectors, for example $\mathbf{p} = (p_x, p_y)$. A dot between two bold quantities is the dot product. t^* is the CPA time. $\boldsymbol{\theta} = [\mathbf{p}; \mathbf{v}; \mathbf{a}]$ is the list of six inputs $[p_x, p_y, v_x, v_y, a_x, a_y]$.

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1 Vectors

1.1 Components and basic operations

A 2D vector is a pair of numbers $\mathbf{a} = (a_x, a_y)$: how far east and how far north. We add and subtract component by component, and multiply by a number (a *scalar*) component by component:

$$\mathbf{a} + \mathbf{b} = (a_x + b_x,\ a_y + b_y), \qquad k\,\mathbf{a} = (ka_x,\ ka_y).$$

Worked example: vector arithmetic

Let $\mathbf{a} = (3, 4)$ and $\mathbf{b} = (-1, 2)$.

$$\mathbf{a} + \mathbf{b} = (3 + (-1),\ 4 + 2) = (2,\ 6)$$
$$\mathbf{a} - \mathbf{b} = (3 - (-1),\ 4 - 2) = (4,\ 2)$$
$$2\mathbf{a} = (2 \cdot 3,\ 2 \cdot 4) = (6,\ 8)$$

1.2 The dot product

$$\mathbf{a}\cdot\mathbf{b} = a_x b_x + a_y b_y.$$

It gives a single number, not a vector. Geometrically $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\phi$, where ϕ is the angle between them. So:

Key idea: perpendicular means zero dot product

$\mathbf{a}\cdot\mathbf{b} = 0$ exactly when the vectors are perpendicular (at right angles). This is the heart of CPA: at the closest point, the relative position is perpendicular to the relative velocity.

Worked example: dot products

$$(3,4)\cdot(-1,2) = 3\cdot(-1) + 4\cdot 2 = -3 + 8 = 5$$
$$(3,4)\cdot(-4,3) = 3\cdot(-4) + 4\cdot 3 = -12 + 12 = 0 \quad \Rightarrow \text{ perpendicular.}$$

Rules you will use (each follows directly from the component definition):

$$\mathbf{a}\cdot\mathbf{b} = \mathbf{b}\cdot\mathbf{a} \qquad \text{(order does not matter)}$$
$$(\mathbf{a} + \mathbf{b})\cdot\mathbf{c} = \mathbf{a}\cdot\mathbf{c} + \mathbf{b}\cdot\mathbf{c} \qquad \text{(distributes over addition)}$$
$$(k\mathbf{a})\cdot\mathbf{b} = k\,(\mathbf{a}\cdot\mathbf{b}) \qquad \text{(scalars come out)}$$

1.3 Length (norm)

The length of a vector is $|\mathbf{a}| = \sqrt{\mathbf{a}\cdot\mathbf{a}} = \sqrt{a_x^2 + a_y^2}$, which is just Pythagoras. In MATLAB, `norm(a)`. Note $|\mathbf{a}|^2 = \mathbf{a}\cdot\mathbf{a}$, which is often easier to work with because it has no square root.

Worked example: length

$|(3,4)| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$

1.4 Bearing and the `atan2` function

Navigation bearing is measured from north ($+y$) toward east ($+x$). For a vector (x, y),

$$\beta = \text{atan2}(x,\, y).$$

(Ordinary math uses atan2(y, x), measured from $+x$. The program's `taninv(xrel,yrel)` follows the navigation convention.) `atan2` is $\arctan(x/y)$ but it also uses the signs of x and y to land in the correct quadrant. Radians convert to degrees by $\text{deg} = \text{rad} \cdot 180/\pi$.

Worked example: bearings

$$\begin{aligned}
(0, 3000) &\to \beta = \text{atan2}(0, 3000) = 0 \text{ rad} = 0^\circ \quad \text{(due north)} \\
(3000, 3000) &\to \beta = \arctan(1) = \pi/4 = 0.7854 \text{ rad} = 45^\circ \\
(3000, 0) &\to \beta = \pi/2 = 90^\circ \quad \text{(due east)} \\
(3147.96,\, 3953.66) &\to \beta = \arctan\big(3147.96/3953.66\big) = \arctan(0.79622) = 0.67243 \text{ rad} \\
&= 0.67243 \cdot \tfrac{180}{\pi} = 38.527^\circ.
\end{aligned}$$

The last vector is the CPA position in our test scenario.

2 Motion and Relative Motion

2.1 Position, velocity, acceleration

Velocity is the rate of change of position, and acceleration is the rate of change of velocity:

$$\mathbf{v} = \frac{d\mathbf{p}}{dt}, \qquad \mathbf{a} = \frac{d\mathbf{v}}{dt}.$$

If the acceleration is *constant*, we can reverse these (integrate). Start with $\dfrac{d\mathbf{v}}{dt} = \mathbf{a}$:

$$\mathbf{v}(t) = \int \mathbf{a}\, dt = \mathbf{a}\, t + C_1.$$

At $t = 0$ the velocity is the starting velocity $\mathbf{v}_0$, so $C_1 = \mathbf{v}_0$ and $\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}\, t$. Now integrate again:

$$\mathbf{p}(t) = \int (\mathbf{v}_0 + \mathbf{a}\, t)\, dt = \mathbf{v}_0\, t + \tfrac{1}{2}\mathbf{a}\, t^2 + C_2.$$

At $t = 0$ the position is $\mathbf{p}_0$, so $C_2 = \mathbf{p}_0$:

$$\boxed{\mathbf{p}(t) = \mathbf{p}_0 + \mathbf{v}_0\, t + \tfrac{1}{2}\, \mathbf{a}\, t^2, \qquad \mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}\, t}$$

Constant velocity is the special case $\mathbf{a} = \mathbf{0}$.

2.2 Relative motion

Let the sensor have $(\mathbf{p}_S, \mathbf{v}_S, \mathbf{a}_S)$ and the contact $(\mathbf{p}_T, \mathbf{v}_T, \mathbf{a}_T)$. The *relative* position is contact minus sensor:

$$\begin{aligned}\mathbf{R}(t) &= \mathbf{p}_T(t) - \mathbf{p}_S(t) \\ &= \left(\mathbf{p}_T + \mathbf{v}_T t + \tfrac{1}{2}\mathbf{a}_T t^2\right) - \left(\mathbf{p}_S + \mathbf{v}_S t + \tfrac{1}{2}\mathbf{a}_S t^2\right) \\ &= (\mathbf{p}_T - \mathbf{p}_S) + (\mathbf{v}_T - \mathbf{v}_S)\, t + \tfrac{1}{2}(\mathbf{a}_T - \mathbf{a}_S)\, t^2.\end{aligned}$$

This has the same form as a single object's motion, with relative initial values $\mathbf{p} = \mathbf{p}_T - \mathbf{p}_S$, $\mathbf{v} = \mathbf{v}_T - \mathbf{v}_S$, $\mathbf{a} = \mathbf{a}_T - \mathbf{a}_S$. This is exactly what the script computes with `Rpos = Tpos - Spos` and so on. So from now on we only need

$$\mathbf{R}(t) = \mathbf{p} + \mathbf{v}\, t + \tfrac{1}{2}\mathbf{a}\, t^2, \qquad \mathbf{V}(t) = \frac{d\mathbf{R}}{dt} = \mathbf{v} + \mathbf{a}\, t.$$

$\mathbf{V}$ is the relative velocity at time t.

Worked example: the test scenario's relative state

Sensor: position $(0, 0)$, velocity $(3, -1)$, acceleration $(0.002,\ 0.003)$. Contact: position $(10000, -5000)$, velocity $(3, 4)$, acceleration $(-0.005,\ 0.005)$. Then

$$\begin{aligned}\mathbf{p} &= (10000 - 0,\ -5000 - 0) = (10000,\ -5000) \\ \mathbf{v} &= (3 - 3,\ 4 - (-1)) = (0,\ 5) \\ \mathbf{a} &= (-0.005 - 0.002,\ 0.005 - 0.003) = (-0.007,\ 0.002).\end{aligned}$$

Check at $t = 100\ s$ using the relative formula:

$$\begin{aligned}\mathbf{R}(100) &= (10000, -5000) + (0, 5)(100) + \tfrac{1}{2}(-0.007, 0.002)(100)^2 \\ &= (10000, -5000) + (0, 500) + (-35,\ 10) = (9965,\ -4490).\end{aligned}$$

Check by tracking each object separately:

$$\begin{aligned}\text{sensor} &= (0,0) + (3,-1)(100) + \tfrac{1}{2}(0.002, 0.003)(10^4) = (300 + 10,\ -100 + 15) = (310,\ -85) \\ \text{contact} &= (10000, -5000) + (3, 4)(100) + \tfrac{1}{2}(-0.005, 0.005)(10^4) = (10275,\ -4575) \\ \text{difference} &= (10275 - 310,\ -4575 - (-85)) = (9965,\ -4490)\ \checkmark\end{aligned}$$

3 Derivative Rules You Need

The derivative $f'(x) = \dfrac{df}{dx}$ is the slope of f: how fast f changes per unit change in x. Formally $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$. The rules below are all we use.

Rule	Formula
Constant	$\dfrac{d}{dx}c = 0$
Power	$\dfrac{d}{dx}x^n = n\,x^{n-1}$
Constant multiple	$\dfrac{d}{dx}\big(c\,f\big) = c\,f'$
Sum	$\dfrac{d}{dx}(f+g) = f' + g'$
Product	$\dfrac{d}{dx}(f\,g) = f'g + fg'$
Chain	$\dfrac{d}{dx}f(u(x)) = f'(u)\,u'(x)$
Quotient	$\dfrac{d}{dx}\dfrac{f}{g} = \dfrac{f'g - fg'}{g^2}$
arctan	$\dfrac{d}{dx}\arctan x = \dfrac{1}{1+x^2}$

Worked example: using the rules

Differentiate $y = 3x^4 - 2x + 7$: $y' = 3\cdot 4x^3 - 2\cdot 1 + 0 = 12x^3 - 2$.
Differentiate $y = (x^2+1)^3$ with the chain rule. Let $u = x^2+1$, so $y = u^3$: $y' = 3u^2\cdot u' = 3(x^2+1)^2\cdot 2x = 6x(x^2+1)^2$.

3.1 Derivatives of dot products

Our key quantities are $\mathbf{R}\cdot\mathbf{R}$ and $\mathbf{R}\cdot\mathbf{V}$, where $\mathbf{R}$ and $\mathbf{V}$ change with time. Write $\mathbf{R} = (R_x, R_y)$ and $\mathbf{V} = (V_x, V_y)$ with $\dfrac{dR_x}{dt} = V_x$ and $\dfrac{dR_y}{dt} = V_y$. Then

$$\begin{aligned}
\frac{d}{dt}(\mathbf{R}\cdot\mathbf{R}) &= \frac{d}{dt}\big(R_x^2 + R_y^2\big) = 2R_xV_x + 2R_yV_y = 2\,\mathbf{R}\cdot\mathbf{V}.\\
\frac{d}{dt}(\mathbf{R}\cdot\mathbf{V}) &= \frac{d}{dt}\big(R_xV_x + R_yV_y\big)\\
&= (V_xV_x + R_x\frac{dV_x}{dt}) + (V_yV_y + R_y\frac{dV_y}{dt}) \quad \text{(product rule)}\\
&= \mathbf{V}\cdot\mathbf{V} + \mathbf{R}\cdot\mathbf{a},
\end{aligned}$$

using $\dfrac{d\mathbf{V}}{dt} = \mathbf{a}$. These two results are used again and again:

$$\boxed{\frac{d}{dt}(\mathbf{R}\cdot\mathbf{R}) = 2\,\mathbf{R}\cdot\mathbf{V}, \qquad \frac{d}{dt}(\mathbf{R}\cdot\mathbf{V}) = \mathbf{V}\cdot\mathbf{V} + \mathbf{R}\cdot\mathbf{a}}$$

3.2 Finding a minimum

A smooth function is at a minimum where its derivative is zero and its second derivative is positive.

Worked example: a minimum

$h(t) = (t-3)^2 + 4$. Then $h'(t) = 2(t-3) = 0 \Rightarrow t = 3$, and $h''(t) = 2 > 0$, so $t = 3$ is a minimum with $h(3) = 4$.

4 Partial Derivatives, Gradients and the Jacobian

4.1 Partial derivatives

When a function has several inputs, a *partial derivative* measures the change caused by one input while all the others are held fixed. Write $\dfrac{\partial g}{\partial x}$: differentiate with respect to x, treat y as a constant.

Worked example: partials of $g(x, y) = x^2 y$

$$\frac{\partial g}{\partial x} = 2xy \qquad \text{(treat } y \text{ as a constant: } \frac{d}{dx}x^2 = 2x\text{)}$$
$$\frac{\partial g}{\partial y} = x^2 \qquad \text{(treat } x \text{ as a constant)}$$

At $(x, y) = (2, 3)$: $\dfrac{\partial g}{\partial x} = 2 \cdot 2 \cdot 3 = 12$ and $\dfrac{\partial g}{\partial y} = 2^2 = 4$. Interpretation: near that point, increasing x by a small δ raises g by about 12δ, and increasing y by δ raises g by about 4δ.

4.2 Total change

If both inputs change a little, the changes add (first-order approximation):

$$\Delta g \approx \frac{\partial g}{\partial x}\,\Delta x + \frac{\partial g}{\partial y}\,\Delta y.$$

Stack the partials into a row (the *gradient*) $\begin{bmatrix}\dfrac{\partial g}{\partial x} & \dfrac{\partial g}{\partial y}\end{bmatrix}$ and this is a matrix product of that row with the column $[\Delta x; \Delta y]$.

4.3 The Jacobian

For several outputs, stack one gradient row per output. This matrix is the *Jacobian*.

Worked example: a small Jacobian

$F(x, y) = \big(xy,\ x + y^2\big)$. Row 1: $\begin{bmatrix}\dfrac{\partial(xy)}{\partial x}, \dfrac{\partial(xy)}{\partial y}\end{bmatrix} = [y,\ x]$. Row 2: $[1,\ 2y]$. So

$$J = \begin{bmatrix} y & x \\ 1 & 2y \end{bmatrix}, \qquad \text{at } (2,3): \ J = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}.$$

Key idea: what "Phi" is in the code

Our problem has three outputs (CPA time, bearing, range) and six inputs $[p_x, p_y, v_x, v_y, a_x, a_y]$. The Jacobian is therefore 3×6. Each row is one of the code's `Phi_T`, `Phi_B`, `Phi_R`. The entry in row i, column j is "how much output i changes per unit change of input j."

5 CPA with Constant Velocity

5.1 Derivation

With $\mathbf{a} = \mathbf{0}$: $\mathbf{R}(t) = \mathbf{p} + \mathbf{v}t$ and $\mathbf{V} = \mathbf{v}$ (constant). The squared range is $r^2(t) = \mathbf{R}\cdot\mathbf{R}$. Its derivative (from Section 3) is

$$\frac{d}{dt} r^2 = 2\,\mathbf{R}\cdot\mathbf{V} = 2(\mathbf{p} + \mathbf{v}t)\cdot\mathbf{v} = 2\big(\mathbf{p}\cdot\mathbf{v} + (\mathbf{v}\cdot\mathbf{v})\,t\big).$$

Set to zero and solve for t:

$$\mathbf{p}\cdot\mathbf{v} + (\mathbf{v}\cdot\mathbf{v})\,t^* = 0 \quad \Longrightarrow \quad \boxed{t^* = -\frac{\mathbf{p}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}}$$

The second derivative of r^2 is $2\,\mathbf{v}\cdot\mathbf{v} > 0$, so this is indeed a minimum. At that time $\mathbf{R}\cdot\mathbf{V} = 0$: *the relative position is perpendicular to the relative velocity.*

Worked example: a complete constant-velocity CPA

$\mathbf{p} = (4000,\ 3000)$ m, $\mathbf{v} = (-4,\ 0)$ m/s.

$$\begin{aligned}
N &= \mathbf{p}\cdot\mathbf{v} = 4000(-4) + 3000(0) = -16000 \\
M &= \mathbf{v}\cdot\mathbf{v} = (-4)^2 + 0^2 = 16 \\
t^* &= -N/M = -(-16000)/16 = 1000 \text{ s} \\
\mathbf{R}^* &= \mathbf{p} + \mathbf{v}\,t^* = (4000 - 4\cdot 1000,\ 3000 + 0) = (0,\ 3000) \\
r^* &= \sqrt{0^2 + 3000^2} = 3000 \text{ m}, \qquad \beta^* = \operatorname{atan2}(0, 3000) = 0.
\end{aligned}$$

Check perpendicularity: $\mathbf{R}^*\cdot\mathbf{v} = 0\cdot(-4) + 3000\cdot 0 = 0$ ✓. (The contact sails west along $y = 3000$ and passes due north of the sensor.)

5.2 Partial derivatives of t^* in the constant-velocity case

Write $t^* = -N/M$ with $N = \mathbf{p}\cdot\mathbf{v}$ and $M = \mathbf{v}\cdot\mathbf{v}$. The needed partials of N and M are

$$\frac{\partial N}{\partial \mathbf{p}} = \mathbf{v}, \quad \frac{\partial N}{\partial \mathbf{v}} = \mathbf{p}, \quad \frac{\partial M}{\partial \mathbf{p}} = \mathbf{0}, \quad \frac{\partial M}{\partial \mathbf{v}} = 2\mathbf{v}.$$

(For instance $N = p_x v_x + p_y v_y$ so $\dfrac{\partial N}{\partial p_x} = v_x$.) By the quotient rule, $\dfrac{\partial t^*}{\partial \theta} = -\dfrac{M\,\dfrac{\partial N}{\partial \theta} - N\,\dfrac{\partial M}{\partial \theta}}{M^2}$. Therefore

$$\frac{\partial t^*}{\partial \mathbf{p}} = -\frac{\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}, \qquad \frac{\partial t^*}{\partial \mathbf{v}} = -\frac{\mathbf{p}}{\mathbf{v}\cdot\mathbf{v}} + \frac{2(\mathbf{p}\cdot\mathbf{v})\,\mathbf{v}}{(\mathbf{v}\cdot\mathbf{v})^2}.$$

Worked example: the numbers

With $N = -16000$, $M = 16$, $\mathbf{v} = (-4, 0)$, $\mathbf{p} = (4000, 3000)$:

$$\frac{\partial t^*}{\partial \mathbf{p}} = -\frac{(-4, 0)}{16} = (0.25,\ 0)$$
$$\frac{\partial t^*}{\partial \mathbf{v}} = -\frac{(4000, 3000)}{16} + \frac{2(-16000)(-4, 0)}{256} = (-250,\ -187.5) + (500,\ 0) = (250,\ -187.5).$$

So the Jacobian row for time is $[0.25,\ 0,\ 250,\ -187.5]$. This matches the existing code's lines `Phi(1,1) = -xrd/rvel2` (which gives $-(-4)/16 = 0.25$) and `Phi(1,3) = (-xr*rvel2 - 2*N*xrd)/rvel2^2`, which with its $N = +16000$ equals $(-4000 \cdot 16 - 2 \cdot 16000 \cdot (-4))/256 = (-64000 + 128000)/256 = 250$. Same answer, same formula.

6 CPA with Acceleration: The Cubic

6.1 Where the cubic comes from

CPA happens when $\mathbf{R}\cdot\mathbf{V} = 0$ (Section 5). Now $\mathbf{R}$ and $\mathbf{V}$ both depend on t, so define

$$f(t) = \mathbf{R}(t)\cdot\mathbf{V}(t) = \big(\mathbf{p} + \mathbf{v}t + \tfrac{1}{2}\mathbf{a}t^2\big)\cdot\big(\mathbf{v} + \mathbf{a}t\big).$$

Multiply out term by term using the distributive rule:

$$\begin{aligned} f(t) = {} & \mathbf{p}\cdot\mathbf{v} + (\mathbf{p}\cdot\mathbf{a})\,t \\ & + (\mathbf{v}\cdot\mathbf{v})\,t + (\mathbf{v}\cdot\mathbf{a})\,t^2 \\ & + \tfrac{1}{2}(\mathbf{a}\cdot\mathbf{v})\,t^2 + \tfrac{1}{2}(\mathbf{a}\cdot\mathbf{a})\,t^3. \end{aligned}$$

Collect like powers of t (and use $\mathbf{v}\cdot\mathbf{a} = \mathbf{a}\cdot\mathbf{v}$):

$$\boxed{f(t) = \tfrac{1}{2}(\mathbf{a}\cdot\mathbf{a})\,t^3 + \tfrac{3}{2}(\mathbf{v}\cdot\mathbf{a})\,t^2 + (\mathbf{v}\cdot\mathbf{v} + \mathbf{p}\cdot\mathbf{a})\,t + \mathbf{p}\cdot\mathbf{v}}$$

This is a *cubic* polynomial: its highest power is t^3. The CPA time is a root of f. Its derivative gives the quantity called D:

$$D = \frac{df}{dt} = \mathbf{V}\cdot\mathbf{V} + \mathbf{R}\cdot\mathbf{a} \qquad \text{(from Section 3)}.$$

Since $f = \frac{1}{2}\,\frac{d}{dt}(\mathbf{R}\cdot\mathbf{R})$, a root of f is a minimum of range when $D > 0$ and a maximum when $D < 0$.

Worked example: the cubic for the test scenario

$\mathbf{p} = (10000, -5000)$, $\mathbf{v} = (0, 5)$, $\mathbf{a} = (-0.007,\ 0.002)$.

$$\begin{aligned}
\mathbf{p}\cdot\mathbf{v} &= 10000 \cdot 0 + (-5000)(5) = -25000 \\
\mathbf{p}\cdot\mathbf{a} &= 10000(-0.007) + (-5000)(0.002) = -70 - 10 = -80 \\
\mathbf{v}\cdot\mathbf{v} &= 0^2 + 5^2 = 25 \\
\mathbf{v}\cdot\mathbf{a} &= 0(-0.007) + 5(0.002) = 0.01 \\
\mathbf{a}\cdot\mathbf{a} &= (-0.007)^2 + (0.002)^2 = 0.000049 + 0.000004 = 0.000053.
\end{aligned}$$

Coefficients: $\frac{1}{2}(0.000053) = 2.65 \times 10^{-5}$; $\frac{3}{2}(0.01) = 0.015$; $25 + (-80) = -55$; -25000. So

$$f(t) = 2.65 \times 10^{-5}\, t^3 + 0.015\, t^2 - 55\, t - 25000.$$

6.2 Solving the cubic: Newton's method

MATLAB's `roots` solves polynomials directly. To see *how* a root can be found, here is Newton's method. Approximate f near a guess t_n by its tangent line, and jump to where that line crosses zero:

$$f(t_n) + f'(t_n)(t_{n+1} - t_n) = 0 \quad \Longrightarrow \quad t_{n+1} = t_n - \frac{f(t_n)}{f'(t_n)}.$$

Here $f'(t) = 3(2.65 \times 10^{-5})t^2 + 2(0.015)t - 55 = 7.95 \times 10^{-5}t^2 + 0.03\, t - 55$.

Worked example: Newton's method, first step in detail

Start at $t_0 = 1000$.

$$\begin{aligned}
f(1000) &= 2.65{\times}10^{-5}(10^9) + 0.015(10^6) - 55(1000) - 25000 \\
&= 26500 + 15000 - 55000 - 25000 = -38500 \\
f'(1000) &= 7.95{\times}10^{-5}(10^6) + 0.03(1000) - 55 = 79.5 + 30 - 55 = 54.5 \\
t_1 &= 1000 - \frac{-38500}{54.5} = 1000 + 706.42 = 1706.42.
\end{aligned}$$

Continuing the same recipe:

n	t_n	$f(t_n)$	$f'(t_n)$
0	1000.0000	-38500.0	54.500
1	1706.4220	56500.5	227.687
2	1458.2719	8872.5	157.809
3	1402.0490	409.17	143.338
4	1399.1944	1.0299	142.617
5	1399.18720	6.6×10^{-6}	142.615

Converged: $t^* = 1399.1872$ s $= 23.320$ min, the value the program prints.

6.3 Which root?

A cubic can have up to three real roots. `roots(c)` for this scenario returns 1399.19, −1522.32 and −442.91. Evaluating $D = \mathbf{V}\cdot\mathbf{V} + \mathbf{R}\cdot\mathbf{a}$ at each:

Root (s)	Range there (m)	D	Meaning
1399.19	5053.8	+142.6	minimum, in the future
−1522.32	10466.0	+83.6	minimum, in the past (farther away)
−442.91	11661.8	−52.7	maximum of range (not a CPA)

My function keeps the roots with $D > 0$ and picks the one with the smallest range, which gives 1399.19. The existing `MLE_compute_cpa_acc` must use a rule that selects the same root here, because both print the same CPA time, bearing and range. Which rule to use (first future minimum? smallest range?) is a design decision, and it matters in other scenarios.

7 Implicit Differentiation: The Key Trick

We need $\dfrac{\partial t^*}{\partial \theta}$, the sensitivity of the CPA time to each input. But t^* is the root of a cubic, and the formula for a cubic's root is enormous. Implicit differentiation avoids it entirely.

7.1 Warm-up examples

Worked example: a circle

$x^2 + y^2 = 1$ defines y as a function of x (without our having to solve for it). Differentiate both sides with respect to x, remembering that y depends on x (chain rule):

$$2x + 2y\,\frac{dy}{dx} = 0 \quad\Longrightarrow\quad \frac{dy}{dx} = -\frac{x}{y}.$$

At (0.6, 0.8): $\dfrac{dy}{dx} = -0.6/0.8 = -0.75$.

Worked example: a root that depends on a parameter

t is the positive root of $f(t, a) = t^2 - a = 0$, so $t = \sqrt{a}$. Differentiate $f(t(a), a) = 0$ with respect to a (chain rule):

$$\frac{\partial f}{\partial t}\,\frac{dt}{da} + \frac{\partial f}{\partial a} = 0 \quad\Longrightarrow\quad \frac{dt}{da} = -\frac{\partial f/\partial a}{\partial f/\partial t} = -\frac{-1}{2t} = \frac{1}{2t}.$$

Check against the direct answer: $\dfrac{d}{da}\sqrt{a} = \frac{1}{2\sqrt{a}} = \frac{1}{2t}$. ✓

7.2 The general rule

Suppose $f(t,\theta)=0$ holds for the root $t=t^*(\theta)$ for every θ. Differentiating $f(t^*(\theta),\theta)=0$ with respect to θ by the chain rule:

$$\frac{\partial f}{\partial t}\frac{\partial t^*}{\partial \theta}+\frac{\partial f}{\partial \theta}=0 \quad\Longrightarrow\quad \boxed{\frac{\partial t^*}{\partial \theta}=-\frac{\partial f/\partial \theta}{\partial f/\partial t}=-\frac{1}{D}\frac{\partial f}{\partial \theta}}$$

All partials on the right are taken *with t held fixed* and are evaluated at $t=t^*$. This is why the method is practical: $\partial f/\partial\theta$ and D are simple.

7.3 Applying it to $f=\mathbf{R}\cdot\mathbf{V}$

First the building blocks, with t fixed. From $\mathbf{R}=\mathbf{p}+\mathbf{v}t+\frac{1}{2}\mathbf{a}t^2$ and $\mathbf{V}=\mathbf{v}+\mathbf{a}t$:

$$\frac{\partial \mathbf{R}}{\partial \mathbf{p}}=I,\quad \frac{\partial \mathbf{R}}{\partial \mathbf{v}}=t\,I,\quad \frac{\partial \mathbf{R}}{\partial \mathbf{a}}=\tfrac{1}{2}t^2 I;\qquad \frac{\partial \mathbf{V}}{\partial \mathbf{p}}=0,\quad \frac{\partial \mathbf{V}}{\partial \mathbf{v}}=I,\quad \frac{\partial \mathbf{V}}{\partial \mathbf{a}}=t\,I,$$

where I is the 2×2 identity matrix. Now use the product rule on $f=\mathbf{R}\cdot\mathbf{V}$:

$$\begin{aligned}\frac{\partial f}{\partial \mathbf{p}}&=I\,\mathbf{V}+0\cdot\mathbf{R}=\mathbf{V}\\ \frac{\partial f}{\partial \mathbf{v}}&=t\,I\,\mathbf{V}+I\,\mathbf{R}=t\mathbf{V}+\mathbf{R}\\ \frac{\partial f}{\partial \mathbf{a}}&=\tfrac{1}{2}t^2\,\mathbf{V}+t\,\mathbf{R}.\end{aligned}$$

Combined with $D=\mathbf{V}\cdot\mathbf{V}+\mathbf{R}\cdot\mathbf{a}$:

$$\boxed{\frac{\partial t^*}{\partial \mathbf{p}}=-\frac{\mathbf{V}^\top}{D},\qquad \frac{\partial t^*}{\partial \mathbf{v}}=-\frac{(t^*\mathbf{V}+\mathbf{R})^\top}{D},\qquad \frac{\partial t^*}{\partial \mathbf{a}}=-\frac{\left(\frac{1}{2}t^{*2}\mathbf{V}+t^*\mathbf{R}\right)^\top}{D}}$$

(Each is a 1×2 row.)

Watch out: sanity check against Section 5

Set $\mathbf{a}=\mathbf{0}$. Then $\mathbf{V}=\mathbf{v}$, $D=\mathbf{v}\cdot\mathbf{v}$, and $\dfrac{\partial t^*}{\partial \mathbf{p}}=-\mathbf{v}/(\mathbf{v}\cdot\mathbf{v})$, exactly the constant-velocity result. In the $\mathbf{v}$ formula, $t^*\mathbf{V}+\mathbf{R}=t^*\mathbf{v}+\mathbf{p}+\mathbf{v}t^*=\mathbf{p}+2t^*\mathbf{v}$, so $\dfrac{\partial t^*}{\partial \mathbf{v}}=-\dfrac{\mathbf{p}+2t^*\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}$; substituting $t^*=-\frac{\mathbf{p}\mathbf{v}}{\mathbf{v}\mathbf{v}}$ gives $-\dfrac{\mathbf{p}}{\mathbf{v}\cdot\mathbf{v}}+\dfrac{2(\mathbf{p}\cdot\mathbf{v})\mathbf{v}}{(\mathbf{v}\cdot\mathbf{v})^2}$, again identical.

7.4 Full numbers for the test scenario

First the state at CPA. With $t^*=1399.1872$ s, $t^{*2}=1\,957\,724.83$ and $\frac{1}{2}t^{*2}=978\,862.417$:

$$\begin{aligned}R_x&=10000+0\cdot t^*+(-0.007)(978862.417)=10000-6852.037=3147.963\\ R_y&=-5000+5(1399.1872)+(0.002)(978862.417)=-5000+6995.936+1957.725=3953.661\\ V_x&=0+(-0.007)(1399.1872)=-9.7943\\ V_y&=5+(0.002)(1399.1872)=5+2.7984=7.7984.\end{aligned}$$

Check $\mathbf{R}\cdot\mathbf{V} = 3147.963(-9.7943) + 3953.661(7.7984) = -30832 + 30832 \approx 0$ ✓. Now

$$\begin{aligned}
\mathbf{V}\cdot\mathbf{V} &= (-9.7943)^2 + (7.7984)^2 = 95.929 + 60.814 = 156.743 \\
\mathbf{R}\cdot\mathbf{a} &= 3147.963(-0.007) + 3953.661(0.002) = -22.036 + 7.907 = -14.128 \\
D &= 156.743 - 14.128 = 142.615.
\end{aligned}$$

The three blocks of $\partial f/\partial\theta$:

$$\begin{aligned}
\frac{\partial f}{\partial \mathbf{p}} &= \mathbf{V} = (-9.7943,\ 7.7984) \\
t^*\mathbf{V} &= (1399.1872\cdot(-9.7943),\ 1399.1872\cdot 7.7984) = (-13704.07,\ 10911.39) \\
\frac{\partial f}{\partial \mathbf{v}} &= t^*\mathbf{V} + \mathbf{R} = (-13704.07 + 3147.96,\ 10911.39 + 3953.66) = (-10556.11,\ 14865.05) \\
\tfrac{1}{2}t^{*2}\mathbf{V} &= 978862.417\cdot(-9.7943,\ 7.7984) = (-9\,587\,282,\ 7\,633\,536) \\
t^*\mathbf{R} &= (1399.1872\cdot 3147.963,\ 1399.1872\cdot 3953.661) = (4\,404\,590,\ 5\,531\,912) \\
\frac{\partial f}{\partial \mathbf{a}} &= (-9587282 + 4404590,\ 7633536 + 5531912) = (-5\,182\,693,\ 13\,165\,447).
\end{aligned}$$

Divide each entry by $-D = -142.615$:

	p_x	p_y	v_x	v_y	a_x	a_y
$\partial f/\partial\theta$	-9.794	7.798	-10556.1	14865.0	-5.183×10^6	1.3165×10^7
$\dfrac{\partial t^*}{\partial\theta} = -\frac{1}{D}\dfrac{\partial f}{\partial\theta}$	0.06868	-0.05468	74.018	-104.232	36340.5	-92314.8

For example $\dfrac{\partial t^*}{\partial p_x} = -(-9.7943)/142.615 = 0.06868$, and $\dfrac{\partial t^*}{\partial a_y} = -13165447/142.615 = -92314.8$. In words: a 0.001 m/s^2 error in a_y moves the CPA time by about -92 s, because the acceleration acts over $t^{*2} \approx 2\times10^6$ s^2.

8 Derivatives of Range and Bearing

8.1 The CPA position depends on the inputs in two ways

$\mathbf{R}^* = \mathbf{R}(t^*(\theta),\theta)$: the inputs enter directly through $\mathbf{p},\mathbf{v},\mathbf{a}$ and indirectly through the time t^* at which we evaluate. The chain rule gives the *total* derivative

$$\frac{d\mathbf{R}^*}{d\theta} = \underbrace{\frac{\partial \mathbf{R}}{\partial\theta}\Big|_t}_{\text{direct, } t \text{ fixed}} + \underbrace{\mathbf{V}\,\frac{\partial t^*}{\partial\theta}}_{\text{via the time shift}},$$

because $\dfrac{\partial \mathbf{R}}{\partial t} = \mathbf{V}$. In matrix form, with $\dfrac{\partial \mathbf{R}}{\partial\theta}\Big|_t = [\,I \;\; tI \;\; \tfrac{1}{2}t^2 I\,]$ (2×6) and $\dfrac{\partial t^*}{\partial\theta}$ a 1×6 row:

$$\frac{d\mathbf{R}^*}{d\boldsymbol{\theta}} = [\,I \;\; t^*I \;\; \tfrac{1}{2}t^{*2}I\,] + \mathbf{V}\,\frac{\partial t^*}{\partial\boldsymbol{\theta}}.$$

8.2 Range: the time term vanishes

Range $r = \sqrt{\mathbf{R}\cdot\mathbf{R}}$. By the chain rule, $\dfrac{dr}{d\theta} = \dfrac{1}{2r}\dfrac{d(\mathbf{R}\cdot\mathbf{R})}{d\theta} = \dfrac{\mathbf{R}}{r}\cdot\dfrac{d\mathbf{R}^*}{d\theta}$. Insert the total derivative:

$$\frac{dr}{d\theta} = \frac{\mathbf{R}^\top}{r}\Big(\frac{\partial\mathbf{R}}{\partial\theta}\Big|_t + \mathbf{V}\frac{\partial t^*}{\partial\theta}\Big) = \frac{\mathbf{R}^\top}{r}\frac{\partial\mathbf{R}}{\partial\theta}\Big|_t + \underbrace{\frac{\mathbf{R}\cdot\mathbf{V}}{r}}_{=0 \text{ at CPA}}\frac{\partial t^*}{\partial\theta}.$$

So the time term drops out:

$$\boxed{\frac{\partial r}{\partial\mathbf{p}} = \frac{\mathbf{R}^\top}{r}, \quad \frac{\partial r}{\partial\mathbf{v}} = t^*\frac{\mathbf{R}^\top}{r}, \quad \frac{\partial r}{\partial\mathbf{a}} = \tfrac{1}{2}t^{*2}\frac{\mathbf{R}^\top}{r}}$$

Intuition: at the closest point the range is, by definition, not changing with time, so a small shift in t^* changes the range only at second order.

Worked example: range partials, test scenario

$r = \sqrt{3147.963^2 + 3953.661^2} = 5053.82$ m, so $\mathbf{R}/r = (0.62289,\ 0.78231)$.

$$\frac{\partial r}{\partial\mathbf{p}} = (0.6229,\ 0.7823)$$
$$\frac{\partial r}{\partial\mathbf{v}} = 1399.1872\,(0.62289,\ 0.78231) = (871.5,\ 1094.6)$$
$$\frac{\partial r}{\partial\mathbf{a}} = 978862.417\,(0.62289,\ 0.78231) = (609\,721,\ 765\,775).$$

8.3 Bearing

$\beta = \operatorname{atan2}(R_x, R_y) = \arctan(R_x/R_y)$ (away from the branch cut). Using $\dfrac{d}{dx}\arctan x = \frac{1}{1+x^2}$ and the chain rule:

$$\frac{\partial\beta}{\partial R_x} = \frac{1}{1 + R_x^2/R_y^2}\cdot\frac{1}{R_y} = \frac{R_y}{R_y^2 + R_x^2} = \frac{R_y}{r^2}$$
$$\frac{\partial\beta}{\partial R_y} = \frac{1}{1 + R_x^2/R_y^2}\cdot\Big(-\frac{R_x}{R_y^2}\Big) = -\frac{R_x}{R_y^2 + R_x^2} = -\frac{R_x}{r^2}.$$

Define the row $\mathbf{g} = \begin{bmatrix} R_y, & -R_x \end{bmatrix}/r^2$. Then

$$\boxed{\frac{d\beta}{d\boldsymbol{\theta}} = \mathbf{g}\Big(\frac{\partial\mathbf{R}}{\partial\boldsymbol{\theta}}\Big|_t + \mathbf{V}\,\frac{\partial t^*}{\partial\boldsymbol{\theta}}\Big) = \underbrace{\mathbf{g}\,\frac{\partial\mathbf{R}}{\partial\boldsymbol{\theta}}\Big|_t}_{\text{fixed-}t^*\text{ part}} + \underbrace{(\mathbf{g}\cdot\mathbf{V})\,\frac{\partial t^*}{\partial\boldsymbol{\theta}}}_{\text{time-shift part}}}$$

Here the time-shift part does *not* vanish, because $\mathbf{g}$ points *along* $\mathbf{V}$ (both are perpendicular to $\mathbf{R}$ at CPA), so $\mathbf{g}\cdot\mathbf{V} \neq 0$. This is the main difference between bearing and range.

Worked example: bearing partials, test scenario, column by column

$r^2 = 25\,541\,105.7$, so $\mathbf{g} = (3953.661, -3147.963)/25541105.7 = (1.54796\times10^{-4},\ -1.23251\times10^{-4})$. Then

$$\begin{aligned}\mathbf{g}\cdot\mathbf{V} &= (1.54796\times10^{-4})(-9.7943) + (-1.23251\times10^{-4})(7.7984)\\ &= -1.5161\times10^{-3} - 9.612\times10^{-4} = -2.4773\times10^{-3}.\end{aligned}$$

Column p_x:

$$\begin{aligned}\text{fixed} &= g_x\cdot 1 = 1.54796\times10^{-4}\\ \text{time-shift} &= (-2.4773\times10^{-3})(0.06868) = -1.7014\times10^{-4}\\ \text{total} &= 1.54796\times10^{-4} - 1.7014\times10^{-4} = -1.534\times10^{-5}\end{aligned}$$

Column a_x:

$$\begin{aligned}\text{fixed} &= g_x\cdot\tfrac{1}{2}t^{*2} = (1.54796\times10^{-4})(978862.4) = 151.52\\ \text{time-shift} &= (-2.4773\times10^{-3})(36340.5) = -90.03\\ \text{total} &= 151.52 - 90.03 = 61.50\end{aligned}$$

Column a_y:

$$\begin{aligned}\text{fixed} &= g_y\cdot 978862.4 = -120.65\\ \text{time-shift} &= (-2.4773\times10^{-3})(-92314.8) = +228.69\\ \text{total} &= -120.65 + 228.69 = 108.04\end{aligned}$$

	p_x	p_y	v_x	v_y	a_x	a_y
fixed-t^* part	1.548×10^{-4}	-1.233×10^{-4}	0.2166	−0.1725	151.52	−120.65
total	-1.53×10^{-5}	1.22×10^{-5}	0.0332	0.0858	61.50	108.04

The two versions differ a lot, even in sign. This is the source of the gap between the existing program's bearing uncertainty (0.1967 rad) and the total-derivative value (0.1247 rad).

8.4 A revealing special case: constant velocity

In the constant-velocity example of Section 5 ($\mathbf{p} = (4000, 3000)$, $\mathbf{v} = (-4, 0)$, $t^* = 1000$, $\mathbf{R}^* = (0, 3000)$, $r = 3000$), the bearing partials are easy to do by hand. $\mathbf{g} = [R_y, -R_x]/r^2 = [3000,\ 0]/9\times10^6 = (1/3000,\ 0)$ and $\mathbf{V} = \mathbf{v} = (-4, 0)$, so $\mathbf{g}\cdot\mathbf{V} = -4/3000 = -1/750$. Using $\dfrac{\partial t^*}{\partial \mathbf{p}} = (0.25, 0)$ and

$\dfrac{\partial t^*}{\partial \mathbf{v}} = (250, -187.5)$ from Section 5:

$$\begin{aligned}
\text{fixed-}t^*\text{ part:}\quad & \frac{\partial \beta}{\partial \mathbf{p}} = \mathbf{g} = (\tfrac{1}{3000}, 0), \quad \frac{\partial \beta}{\partial \mathbf{v}} = t^*\,\mathbf{g} = (\tfrac{1}{3},\ 0) \\
\text{time-shift part:}\quad & (\mathbf{g}\cdot\mathbf{V})\frac{\partial t^*}{\partial \mathbf{p}} = -\tfrac{1}{750}(0.25, 0) = (-\tfrac{1}{3000},\ 0) \\
& (\mathbf{g}\cdot\mathbf{V})\frac{\partial t^*}{\partial \mathbf{v}} = -\tfrac{1}{750}(250, -187.5) = (-\tfrac{1}{3},\ 0.25) \\
\text{total:}\quad & \frac{\partial \beta}{\partial \mathbf{p}} = (0,\ 0), \qquad \frac{\partial \beta}{\partial \mathbf{v}} = (0,\ 0.25).
\end{aligned}$$

Key idea: in the constant-velocity case the position cannot affect the CPA bearing

At CPA the relative position is the part of $\mathbf{p}$ perpendicular to $\mathbf{v}$, so the CPA bearing is just "90° off the relative velocity direction" and does not depend on where along the track you start. The total derivative says so exactly ($\dfrac{\partial \beta}{\partial \mathbf{p}} = 0$). The fixed-$t^*$ shortcut gets this wrong: it gives a nonzero position sensitivity $\frac{1}{3000}$ per metre. The velocity result also makes sense: if v_y changes by ϵ, the velocity direction turns by $\epsilon/|\mathbf{v}| = \epsilon/4$, so $\dfrac{\partial \beta}{\partial v_y} = 0.25$. ✓

For this example with position std 100 m and velocity std 0.1 m/s: the fixed-t^* bearing std is $\sqrt{(\frac{100}{3000})^2 + (\frac{0.1}{3})^2} = \sqrt{0.001111 + 0.001111} = 0.0471$ rad, while the total derivative gives $\sqrt{(0.25 \cdot 0.1)^2} = 0.0250$ rad. The shortcut overstates the bearing uncertainty by almost a factor of two.

9 Error Propagation

9.1 Variance and standard deviation

An uncertain input x is modelled as its true value plus a random error with average zero. The *standard deviation* σ_x measures the typical size of that error, and the *variance* is σ_x^2. The covariance matrix $\hat{P}$ in the program lists the variances of the six inputs on its diagonal:

$$\hat{P} = \operatorname{diag}\left(100^2,\ 100^2,\ 0.1^2,\ 0.1^2,\ 0.001^2,\ 0.001^2\right),$$

meaning position errors of about 100 m, velocity errors of about 0.1 m/s, and acceleration errors of about 0.001 m/s^2. A diagonal matrix says the six errors are independent.

9.2 Two basic facts

1. **Scaling.** Multiplying a quantity by a constant k multiplies its standard deviation by $|k|$: $\sigma_{kX} = |k|\,\sigma_X$, so the variance is multiplied by k^2.
2. **Independent errors add in variance.** If X and Y have independent errors, $\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2$ (not the standard deviations themselves).

Worked example: combining two errors

$\sigma_X = 3$ and $\sigma_Y = 4$, independent. Then $\sigma^2_{X+Y} = 9 + 16 = 25$ so $\sigma_{X+Y} = 5$, not $3 + 4 = 7$.

9.3 One input: linearization

For a smooth function g, a small input error δ produces

$$g(x + \delta) \approx g(x) + g'(x)\,\delta.$$

By the scaling rule, $\sigma_g \approx |g'(x)|\,\sigma_x$.

Worked example: one variable

$g(x) = x^2$, $x = 10 \pm 0.1$. $g'(10) = 2 \cdot 10 = 20$, so $\sigma_g \approx 20 \cdot 0.1 = 2.0$. Exact check: $(10.1)^2 - 10^2 = 2.01$. ✓

9.4 Several independent inputs

Each input contributes $\left(\frac{\partial g}{\partial x_k}\right)\delta_k$ to the output error. These contributions add, and since the errors are independent their variances add, giving

$$\sigma_g^2 \approx \sum_k \left(\frac{\partial g}{\partial x_k}\right)^2 \sigma_k^2.$$

Worked example: two variables

$g(x, y) = x^2 + 3y$, with $x = 2 \pm 0.1$ and $y = 5 \pm 0.2$. Partials: $\dfrac{\partial g}{\partial x} = 2x = 4$, $\dfrac{\partial g}{\partial y} = 3$.

$$\sigma_g^2 = 4^2(0.1)^2 + 3^2(0.2)^2 = 16(0.01) + 9(0.04) = 0.16 + 0.36 = 0.52, \qquad \sigma_g = \sqrt{0.52} = 0.721.$$

9.5 The matrix form: $J\hat{P}J^\top$

Collect the input errors in a column $\boldsymbol{\delta}$ and the partials of one output in a row J. The output error is $\Delta g \approx J\boldsymbol{\delta}$. Its variance is the average of $(J\boldsymbol{\delta})(J\boldsymbol{\delta})^\top = J\,\boldsymbol{\delta}\boldsymbol{\delta}^\top J^\top$, and the average of $\boldsymbol{\delta}\boldsymbol{\delta}^\top$ is by definition the covariance matrix $\hat{P}$. Hence

$$\boxed{\sigma_g^2 = J\,\hat{P}\,J^\top}$$

This is exactly the code line `sqrt(Phi_T*Phat*Phi_T')`. It is more general than the sum above because $\hat{P}$ may have nonzero off-diagonal entries (correlated errors).

9.6 Matrix multiplication refresher

For a row J $(1 \times n)$ times a matrix P $(n \times n)$: entry j of JP is $\sum_i J_i P_{ij}$. Then multiply that row by the column $J^\top$ to get a single number.

Worked example: correlated vs independent

$J = [2\ \ 3]$.

Correlated: $P = \begin{bmatrix} 4 & 1 \\ 1 & 9 \end{bmatrix}$.

$$JP = [\,2 \cdot 4 + 3 \cdot 1,\ \ 2 \cdot 1 + 3 \cdot 9\,] = [\,11,\ 29\,]$$
$$JPJ^\top = 11 \cdot 2 + 29 \cdot 3 = 22 + 87 = 109.$$

Independent (diagonal): $P = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$. Then $JP = [8,\ 27]$ and $JPJ^\top = 8 \cdot 2 + 27 \cdot 3 = 16 + 81 = 97$, which equals $J_1^2 P_{11} + J_2^2 P_{22} = 4 \cdot 4 + 9 \cdot 9 = 97$. ✓ For a diagonal P the matrix product reduces to the simple sum.

For all three outputs at once, $\Sigma = J\hat{P}J^\top$ with J being 3×6 gives a 3×3 matrix. Its diagonal holds the variances $\sigma_t^2, \sigma_\beta^2, \sigma_r^2$. The off-diagonal entries are the covariances between outputs, which the program does not use.

9.7 Units

Every term must come out in the output's squared unit. For time, $\dfrac{\partial t^*}{\partial a_y}$ has units s per (m/s^2) and σ_{a_y} has units m/s^2, so the product is in seconds. Bearing is in radians, so the printed value in degrees is a conversion at the very end.

9.8 When the linear approximation is trustworthy

The formulas assume the output is close to a straight line over a range of a few σ around the true inputs. When an input error is large relative to how quickly the function curves, linearization degrades. Here the acceleration errors are small in absolute terms (0.001) but the output curves quickly with them, which is why numerical derivatives with "innocent" step sizes can go wrong (Section 11).

10 The Full Scenario, End to End

Inputs: $\boldsymbol{\theta} = [10000, -5000, 0, 5, -0.007, 0.002]$; $\sigma = [100, 100, 0.1, 0.1, 0.001, 0.001]$. CPA: $t^* = 1399.1872$ s, $\mathbf{R}^* = (3147.963, 3953.661)$, $\mathbf{V} = (-9.7943, 7.7984)$, $r = 5053.82$ m, $\beta = 0.67243$ rad, $D = 142.615$.

For each output, the variance is the sum over inputs of $(J_k\sigma_k)^2$. Worked entries:

$$\text{time, } a_y : (-92314.8 \cdot 0.001)^2 = (-92.3148)^2 = 8522.0$$
$$\text{time, } v_x : (74.018 \cdot 0.1)^2 = (7.4018)^2 = 54.79$$
$$\text{range, } p_x : (0.62289 \cdot 100)^2 = 3879.9.$$

Full tables of $(J_k\sigma_k)^2$:

	p_x	p_y	v_x	v_y	a_x	a_y	**sum**
Time (s^2)	47.16	29.90	54.79	108.64	1320.63	8522.02	**10083.15**
Bearing, total (rad^2)	2.4×10^{-6}	1.5×10^{-6}	1.10×10^{-5}	7.35×10^{-5}	0.003782	0.011673	**0.015544**
Bearing, fixed-t^*	0.000240	0.000152	0.000469	0.000297	0.022960	0.014555	**0.038673**
Range (m^2)	3879.9	6120.1	7595.8	11981.5	371760.2	586411.4	**987748.9**

Square roots:

	Standard deviation	In friendlier units	Existing program
Time	$\sqrt{10083.15} = 100.415$ s	1.674 min	97.449 s (1.624 min)
Bearing, total	$\sqrt{0.015544} = 0.12468$ rad	7.144°	–
Bearing, fixed-t^*	$\sqrt{0.038673} = 0.19665$ rad	11.267°	0.196654 rad (11.267°)
Range	$\sqrt{987748.9} = 993.856$ m	0.994 km	993.856 m

Key idea: what the tables teach

The acceleration errors dominate everything: a_y alone supplies $8522/10083 = 84\%$ of the time variance, and a_x another 13%. Position and velocity errors barely matter for time. So the *accuracy of the partial derivatives with respect to acceleration* decides whether the uncertainty is right, and these are exactly the ones that are hardest to get numerically (next section).

11 Numerical Differentiation and Why the Step Size Matters

11.1 Forward and central differences

Replace the limit in the derivative with a small but finite step h:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \quad \text{(forward)}, \qquad f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \quad \text{(central)}.$$

How wrong are they? Use the Taylor series $f(x \pm h) = f \pm hf' + \frac{h^2}{2}f'' \pm \frac{h^3}{6}f''' + \cdots$.

$$\frac{f(x+h) - f(x)}{h} = f' + \tfrac{h}{2}f'' + \tfrac{h^2}{6}f''' + \cdots \qquad \Rightarrow \text{error} \approx \tfrac{h}{2}f''$$

$$\frac{f(x+h) - f(x-h)}{2h} = f' + \tfrac{h^2}{6}f''' + \cdots \qquad \Rightarrow \text{error} \approx \tfrac{h^2}{6}f'''$$

(for the central one, subtract the two series: the f and f'' terms cancel). The forward error shrinks like h, the central error like h^2, so central differences are much better for the same step.

Worked example: $f(x) = x^3$ at $x = 2$, true derivative $3x^2 = 12$

h	forward	central
1	19	13
0.1	12.61	12.01
0.01	12.0601	12.0001
0.001	12.006001	12.000001

For $h = 0.1$: forward $= (2.1^3 - 2^3)/0.1 = (9.261 - 8)/0.1 = 12.61$; central $= (2.1^3 - 1.9^3)/0.2 = (9.261 - 6.859)/0.2 = 12.01$. Here $f'' = 6x = 12$, so the forward error $\approx \frac{h}{2} \cdot 12 = 0.6$ matches the observed 0.61.

11.2 The opposite danger: steps that are too small

Computers store about 16 digits. If h is so small that $f(x+h)$ and $f(x)$ agree in most of their digits, subtracting them destroys the remaining digits (*cancellation*), and dividing by tiny h magnifies the noise. So there is a sweet spot: not too big (truncation error), not too tiny (rounding error). For central differences a relative step near 10^{-6} to 10^{-5} of the variable's size is a common starting point.

11.3 Our problem: the CPA time is very curved in the accelerations

Compare numerical slopes of t^* with respect to a_y and a_x (obtained by actually re-solving the cubic) against the exact algebraic values -92314.77 and 36340.51:

Step h (m/s^2)	slope w.r.t. a_y		slope w.r.t. a_x	
	forward	central	forward	central
10^{-6}	-92313.97	-92314.77	36337.67	36340.51
10^{-4}	-92223.65	-92304.05	36046.67	36331.28
10^{-3}	-90584.54	-91240.17	32405.78	35400.04
3×10^{-3}	$-1\,080\,261$	$-580\,870$	13754.12	26604.47

Two lessons.

1. **Step size comparable to the uncertainty is a trap.** Because $\sigma_a = 0.001$, a step of 10^{-3} is a full standard deviation, and t^* is far from linear over that span. At $h = 10^{-3}$ the forward slope for a_x is already 10.8% low (32405.78 versus 36340.51). Since a_x and a_y supply almost all the variance (Section 10), the time uncertainty comes out too low. A forward difference with $h \approx 10^{-3}$ on all six inputs reproduces the existing program's 97.449 s.

2. **Large steps can switch roots.** At $h = 3 \times 10^{-3}$ the $a_y + h$ re-solve returns $t^* = -1841.6$ s: the solver picked a different root, so the "slope" is nonsense (-10^6). Algebraic partials have no step to choose and cannot jump roots.

Key idea: how we used numerical differences well

A *tiny central step* (about 10^{-6} relative) is excellent as a *check* on algebra, because both it and the algebra then agree to 6 digits. That is what we did: the finite difference of the algebraic function matched it exactly (100.415 and 0.124675). It is a poor *production method* here, because the right step depends on the scenario.

12 Practice Problems with Full Solutions

Practice problem 1: vectors

$\mathbf{a} = (5, 12)$, $\mathbf{b} = (-12, 5)$. Find $\mathbf{a}\cdot\mathbf{b}$, $|\mathbf{a}|$, whether they are perpendicular, and $\mathbf{a} + 2\mathbf{b}$.

Solution. $\mathbf{a}\cdot\mathbf{b} = 5(-12) + 12(5) = -60 + 60 = 0$, so they are perpendicular. $|\mathbf{a}| = \sqrt{25 + 144} = \sqrt{169} = 13$. $2\mathbf{b} = (-24, 10)$, so $\mathbf{a} + 2\mathbf{b} = (5 - 24,\ 12 + 10) = (-19,\ 22)$.

Practice problem 2: constant-velocity CPA

$\mathbf{p} = (6000, 0)$ m, $\mathbf{v} = (-2, 2)$ m/s. Find t^*, $\mathbf{R}^*$, range, bearing, and $\dfrac{\partial t^*}{\partial p_x}$, $\dfrac{\partial t^*}{\partial p_y}$.

Solution. $N = \mathbf{p}\cdot\mathbf{v} = 6000(-2) + 0 \cdot 2 = -12000$; $M = \mathbf{v}\cdot\mathbf{v} = 4 + 4 = 8$. $t^* = -N/M = 12000/8 = 1500$ s. $\mathbf{R}^* = (6000 - 2 \cdot 1500,\ 0 + 2 \cdot 1500) = (3000,\ 3000)$. Range $= \sqrt{3000^2 + 3000^2} = 3000\sqrt{2} = 4242.64$ m. Bearing $= \text{atan2}(3000, 3000) = \pi/4 = 45°$. Perpendicularity check: $\mathbf{R}^*\cdot\mathbf{v} = 3000(-2) + 3000(2) = 0$ ✓. Partials: $\dfrac{\partial t^*}{\partial \mathbf{p}} = -\mathbf{v}/M = -(-2, 2)/8 = (0.25,\ -0.25)$, so $\dfrac{\partial t^*}{\partial p_x} = 0.25$ and $\dfrac{\partial t^*}{\partial p_y} = -0.25$.

Practice problem 3: implicit differentiation

t solves $f(t, a) = t^3 + at - 10 = 0$. At $a = 1$ the root is $t = 2$. Find $\dfrac{dt}{da}$ there.

Solution. Check: $8 + 2 - 10 = 0$ ✓. $\dfrac{\partial f}{\partial t} = 3t^2 + a = 12 + 1 = 13$. $\dfrac{\partial f}{\partial a} = t = 2$. So $\dfrac{dt}{da} = -\dfrac{2}{13} = -0.1538$. (Numerical check: solving at $a = 1.001$ gives $t \approx 1.99985$, slope ≈ -0.154.)

Practice problem 4: error propagation

$g = xy$ with $x = 10 \pm 0.5$, $y = 4 \pm 0.2$, independent. Find σ_g.

Solution. $\dfrac{\partial g}{\partial x} = y = 4$, $\dfrac{\partial g}{\partial y} = x = 10$. $\sigma_g^2 = 4^2(0.5)^2 + 10^2(0.2)^2 = 16(0.25) + 100(0.04) = 4 + 4 = 8$, so $\sigma_g = \sqrt{8} = 2.83$.

Practice problem 5: bearing partials

$\mathbf{R} = (3, 4)$ with independent errors $\sigma_{R_x} = \sigma_{R_y} = 0.5$. Find β and σ_β.

Solution. $\beta = \text{atan2}(3, 4) = \arctan(0.75) = 0.6435$ rad $= 36.87°$. $r^2 = 25$. $\dfrac{\partial \beta}{\partial R_x} = R_y/r^2 =$

$4/25 = 0.16$; $\dfrac{\partial \beta}{\partial R_y} = -R_x/r^2 = -3/25 = -0.12$. $\sigma_\beta^2 = (0.16 \cdot 0.5)^2 + (0.12 \cdot 0.5)^2 = 0.0064 + 0.0036 = 0.01$, so $\sigma_\beta = 0.1$ rad (5.73°).

Practice problem 6: Newton's method

Find a root of $f(t) = t^2 - 2$ starting from $t_0 = 1$ (two steps).

Solution. $f'(t) = 2t$. $t_1 = 1 - \dfrac{1-2}{2} = 1 + 0.5 = 1.5$. $f(1.5) = 0.25$, $f'(1.5) = 3$, so $t_2 = 1.5 - \dfrac{0.25}{3} = 1.41667$. The true root $\sqrt{2} = 1.41421$.

Practice problem 7: why correlation matters

$J = [1 \quad -1]$, $P = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Compute $JPJ^\top$, and compare with the value obtained by ignoring the off-diagonal terms.

Solution. $JP = [\,1 \cdot 4 + (-1) \cdot 2, \;\; 1 \cdot 2 + (-1) \cdot 9\,] = [\,2, \; -7\,]$. $JPJ^\top = 2 \cdot 1 + (-7)(-1) = 9$. Ignoring correlation: $4 + 9 = 13$. The positive correlation cancels part of the difference $x_1 - x_2$ (Var $= 4 + 9 - 2 \cdot 2 = 9$).

13 Math-to-MATLAB Dictionary

Math	Variable in code	Meaning
$\mathbf{p}, \mathbf{v}, \mathbf{a}$	`p, v, a` (`Rpos, Rvel, Racc`)	relative position, velocity, acceleration
Cubic coefficients	`c`	$[\frac{1}{2}\mathbf{a}\cdot\mathbf{a},\ \frac{3}{2}\mathbf{v}\cdot\mathbf{a},\ \mathbf{v}\cdot\mathbf{v}+\mathbf{p}\cdot\mathbf{a},\ \mathbf{p}\cdot\mathbf{v}]$
t^*	`t` (`CPA_time`)	CPA time
$\mathbf{R}^*,\ \mathbf{V}$	`R, V`	relative position, velocity at CPA
r	`r` (`CPA_rng`)	range at CPA
$D = \mathbf{V}\cdot\mathbf{V} + \mathbf{R}\cdot\mathbf{a}$	`D`	slope of f at the root
$\partial\mathbf{R}/\partial\boldsymbol{\theta}\|_t$	`dRdth`	$[I \ \ tI \ \ \frac{1}{2}t^2 I]$, 2×6
$\partial f/\partial\boldsymbol{\theta}\|_t$	`dfdth`	$[\mathbf{V}^\top \ \ (t\mathbf{V}+\mathbf{R})^\top \ \ (\frac{1}{2}t^2\mathbf{V}+t\mathbf{R})^\top]$
$\partial t^*/\partial\boldsymbol{\theta}$	`dtdth` (`Phi_T`)	$-$`dfdth`$/D$
$d\mathbf{R}^*/d\boldsymbol{\theta}$	`dRtot`	`dRdth + V*dtdth`
$\mathbf{g}$	`[R(2), -R(1)]/r^2`	bearing row
$\partial r/\partial\boldsymbol{\theta}$	`(R'/r)*dRtot`	range row (equal to `(R'/r)*dRdth`)
J	`J`	3×6 Jacobian: time, bearing, range
$\hat{P}$	`Phat`	6×6 covariance
$\sqrt{J\hat{P}J^\top}$	`sqrt(S(i,i))`, `S = J*Phat*J'`	standard deviations

14 Common Mistakes and a Self-Check Recipe

1. **Radians versus degrees.** Derivatives of angles are per radian. Convert only at the end (`*180/pi`).
2. `atan2` **argument order.** The navigation bearing is atan2(x, y), not (y, x).
3. **Forgetting that t^* depends on the inputs.** Holding it fixed is correct for range but not for bearing (Section 8).
4. **Mixing up the state order.** $\hat{P}$ must follow the same order as the Jacobian columns: $[p_x\, p_y\, v_x\, v_y\, a_x\, a_y]$.
5. **Using standard deviations where variances belong.** Only variances add.
6. **Trusting one finite-difference step.** Always try two or three steps, and compare.

A recipe to check any derivative you derive. (1) Compute the quantity at your test point. (2) Nudge one input by a tiny amount, about 10^{-6} of its size, in both directions and re-compute. (3) Divide the difference by twice the nudge. (4) Compare with your formula; they should agree

to at least 5 digits. (5) Repeat for each input column. If one column disagrees, the error is in that column's formula, which tells you where to look.

A suggested study order. Sections 1–3 (an evening), then 4 and 9 (an evening each), then 5 and 6, then 7 and 8 (the core), and finally 10 and 11 to tie everything back to the program. Work the practice problems in Section 12 with the answers covered.